

## Monitoring of Storage Unit in the Grid Structure

SITE : \$.SiteId  
 OPERATOR : \$.SiteOperator  
 MONITORING PERIOD : \$.MoniPeriod  
 EVALUATION DATE : \$.EvaDate

### Introduction

This report provides analysis and presentation of measured data for the storage system

\$.SiteId

SiteDescription

The storage unit is located at

It enables installation and operation of an enlarged PV-Generator with 2.5 times higher peak power by means of buffering PV generation, ensuring the line power to stay within the limits allowed.

The report is conforming with the EC Guidelines for unified monitoring and performance assessment of storage-units in the grid structure.

Thus, it reveals energy balancing in a subgrid, characterized by both energy surplus and autonomy ratio. Further it quantifies and details contribution of the storage unit to grid control and balancing, and thus relief of the utility interface, resp. power line.

In the form of directed power-flows, monitoring data is suitable for spatial aggregation with results from other storage units in a common grid area.

Further to reporting for storage and distribution system operators, guideline conformant data and monitoring evaluation qualify for sharing performance experience, e.g. in the JRC's storage observatory.

### General Data

|                   |                                  |                                      |
|-------------------|----------------------------------|--------------------------------------|
| Storage Unit      | Nominal Power/[MW]: \$.NomStoPow | Nominal Capacity/[MWh]: \$.NomStoCap |
| Utility Interface | Nominal Power/[MW]: \$.NomUIPPow | Site : \$.SiteId                     |

## Energetic Balances

In line with recording format from independent metering, the total energies in the monitoring period sum up to :

|         | Input |     | Output |     |
|---------|-------|-----|--------|-----|
| Subgrid | 17.5  | MWh | 19.5   | MWh |
| Utility | 30.0  | MWh | 30.0   | MWh |
| Storage | 28.0  | MWh | 25.0   | MWh |

with the total difference between in- and outputs covering the storage losses.

In accumulation of directed powerflows, the energy exchanges of the storage unit with subgrid surpluses(net generation) and deficits(net-load), and with the utility can be distinguished from the direct flows between subgrid and utility.

|                             |       |     |
|-----------------------------|-------|-----|
| net-generation to utility   | 19.58 | MWh |
| storage from net-generation | 25.00 | MWh |
| storage from utility        | 3.21  | MWh |
| storage to utility          | 2.73  | MWh |
| storage to net-load         | 25.07 | MWh |
| net-load from utility       | 30.84 | MWh |

## Exchange Yields

Normalizing the directed powerflows, to the nominal power of the storage unit, is resulting in yields, comparable for units of different sizes. Chart 1 shows the exchange yields in their daily averages .

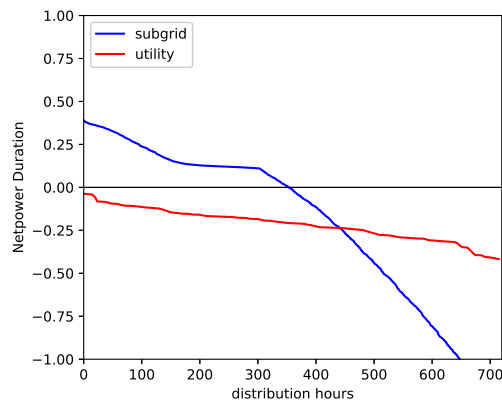


chart 1 : Average DailyYields

Accordingly chart 2 shows the hourly averages of the exchange yields, reflecting average operation in the daily course.

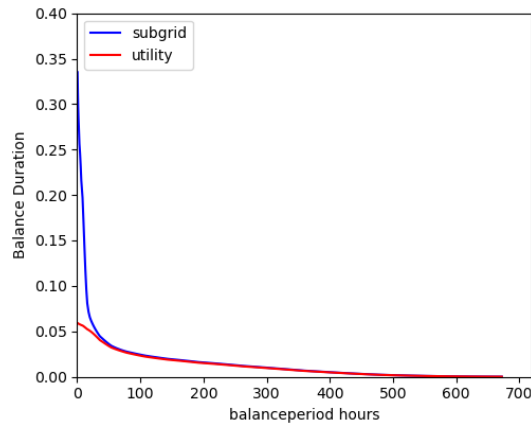


chart 2 : Average Hourly Yields

## Storage Performance in the Grid

Performance is resulting from average yields in the monitoring period, indicating the contribution of the storage unit to balancing. control. This compares to the total balancing need of the subgrid and the remaining balancing control, required from the utility interface.

### Subgrid Yields

Input : \_\_\_\_ Output : \_\_\_\_ Balance Depth : 10.5 hrs

### Utility Yields

Input : \_\_\_\_ Output : \_\_\_\_ Balance Depth : 17.0 hrs

### Storage Yields

Input : \_\_\_\_ Throughput : \_\_\_\_ Balance Depth : 23.0 hrs

Subgrid Balancing \_\_\_\_ 20.5 hrs

Shifted Generation \_\_\_\_ 2.0 hrs

Shifted Load Coverage \_\_\_\_ 0.5 hrs

Operation of the storage unit in time-shift of generation surplus, resp. of load deficit, is separated from operation in balancing the subgrid generation surplus with its load deficit. Time-shift operation reflects operation control performance and includes all exchanges with the utility interface, as well as planned services as unplanned. The storage usage and the longterm energetic losses are characterized by

Recharge Ratio : 112 % Equivalent Full-Cycles : 120

## Utility Interface Relief

Further resolving the balancing needs over increasing timespans, resp. balancing periods, results in the Balance Duration Curve of throughput, and the corresponding capacity as integral over the periods, visible as area under the curves.

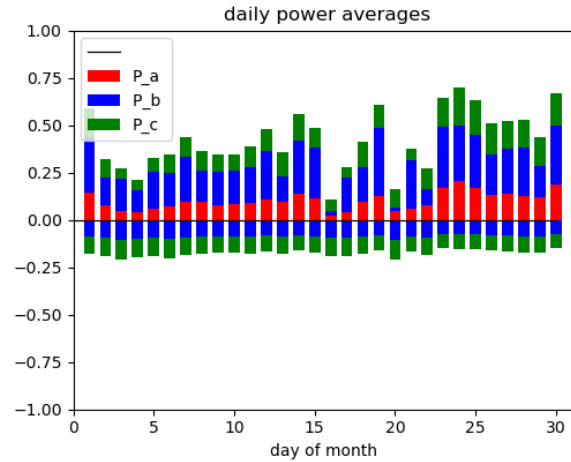


figure 1 : Yields vs Balance-Durations before and behind Storage

With the balancing yields of the subgrid in red, storage is balancing net-power yields over the red area for the balancing capacity employed, the normalized balance depth in [hrs]. The residual utility balance duration curve in blue remains to be compensated by the outer grid infrastructure., with the blue area for the total balancing depth required. Complementary to the energetic balancing, the effect of storage unit operation on releasing the grid interface from power amplitude is quantified. This graph reveals how storage operation is modifying the characteristics of normalized power yields. For the subgrid yields in red, allocated storage is releasing the utility interface, which then covers the blue part only.

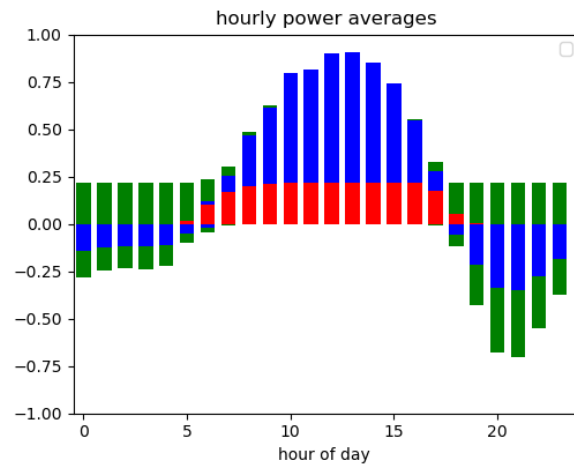


figure 2 : Yields vs Power-Durations before and behind Storage

## Subgrid Coverage Performance Ratios

In relation to the average net load yield  $\bar{Y}_{NL}$  in the monitoring period, subgrid coverage is characterized by performance ratios. Storage and utility are covering the total net load of the subgrid

|                | Net-Surplus Ratio | Throughput Ratio | Net-Load Coverage |
|----------------|-------------------|------------------|-------------------|
| <b>Subgrid</b> | 120 %             | 100 %            | 21 hrs            |
| Utility        | 50 %              | 30 %             | 14.5 hrs          |
| Storage        | 70 %              | 70 %             | 6.5 hrs           |

## Annuities

Based on storage unit power cost of :- PowCost -: Euro per MW-: , and capacity\_cost of :- CapCost -: Euro per MWh on a financing term of :- FinTerm -: years and :- OpEx -:% operation costs, the annuities are resulting as:

| Total expenditure annuity | power | capacity | operation |
|---------------------------|-------|----------|-----------|
| 23 Euro/MWh               | 32%   | 60%      | 8%        |